Classifying and Predicting Degree Trajectories in Longitudinal Ego Networks

Omar Lizardo[*] **David Hachen**[†]

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Abstract

Personal ego networks undergo substantial restructuring during major life transitions, yet research often treats personal network size (degree) as a static individual trait. Drawing on eight-wave panel data from the *NetHealth* Study tracking an undergraduate cohort from matriculation in August 2015 to senior graduation in May 2019 ($N = 457$), we investigate whether individuals follow distinct, predictable degree pathways across the collegiate life course and examine the baseline sociodemographic and psychological determinants of these trajectories. Overcoming the limitations of heuristic clustering and arbitrary imputation, we implement Latent Class Growth Analysis (LCGA) with repeated-measures Poisson finite mixture models across all eight semesters. Model selection identifies three distinct developmental trajectory classes: a stable cadre of "Network Conservers" (28.7%) who maintain large personal networks ($\bar{D} \approx$ 17–19 alters) throughout college; a modal group of "Moderate Winnowers" (39.8%) whose social circles contract steadily from 14.5 to 9.0 alters; and an "Accelerated Winnower" class (31.5%) that undergoes an aggressive early winnowing from 11.0 down to an intimate core of 4.8 alters. Multinomial logistic regressions reveal that trajectory group membership is governed primarily by baseline psychological dispositions (Extraversion, Generalized Trust, Openness) and race rather than parental socioeconomic status. Crucially, decomposing degree across four functional relational layers demonstrates that the aggregate contraction of collegiate networks represents the adaptive pruning of superficial campus acquaintances rather than an erosion of core personal communities: daily activated ties (4.4–5.9 alters) and close, support-providing ties (10.5–12.0 alters) remain invariant across all four years.

[*]Department of Sociology, University of California, Los Angeles. Email: olizardo@soc.ucla.edu.
[†]Department of Sociology, University of Notre Dame. Email: dhachen@nd.edu.

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1 Introduction

Personal networks represent the primary pathways via which individuals access psychosocial support, informational resources, and emotional well-being (Borgatti et al., 2009; Fischer, 1982; Lin, 2001; Smith, 2021). Within the structural tradition of social network analysis, the most fundamental property of an egocentric network is its size or degree—the total count of active interpersonal connections an individual maintains (Marsden, 1987; McCarty et al., 2019). While classical cross-sectional studies documented substantial inter-individual variance in personal network size (Campbell and Lee, 1991; Marsden, 1987), theoretical accounts of network evolution have increasingly emphasized that personal degree is not a static, invariant parameter. Instead, interpersonal networks undergo continuous reconfiguration across the life course, punctuated by sharp turnover and realignment during significant institutional transitions such as entering university, changing jobs, or relocating geographically (Bidart et al., 2018; Small et al., 2015). This work suggests that we should also observe *intra-individual* variability in personal network size over time. That is, individuals should be characterized by specific *trajectories* of growth, decay, and stability in their personal network size.

Despite wide consensus that personal networks evolve dynamically, empirical inquiry into degree trajectories has faced persistent methodological and substantive challenges. On the one hand, evolutionary and cognitive models of human sociality—most notably Dunbar (1992, 1998) social brain hypothesis and recent formulations from complex systems researchers pointing to specific social signatures in the way individuals distribute attention across ties (Heydari et al., 2018; Saramäki et al., 2014)—posit that cognitive processing limits, emotional bandwidth, and temporal budgets impose strict upper bounds on personal network volume. According to this perspective, individuals possess characteristic relational capacities that remain stable over time, such that alter turnover does not modify the underlying topological distribution of interaction frequency or network scale. On the other hand, network analysts in sociology emphasize that relational opportunities are heavily structured by institutional foci, residential arrangements, and organizational contexts (Feld, 1982; Small, 2009; Wang et al., 2020). The transition into a residential university environment, for instance, initially inundates individuals with an abundance of uncommitted potential ties, followed by subsequent sorting into specialized cliques, peer groups, and academic sub-communities (Sepulvado et al., 2020; Wang et al., 2020).

In a foundational preliminary study presented at the Sunbelt network conference, Chandler and Hachen (2018) initiated the empirical investigation of degree trajectories by applying deductive decision trees and heuristic k-means clustering to the first six survey waves of the NetHealth study. Their exploratory analysis yielded two vital initial insights: first, student ego networks exhibit pronounced intra-individual variability over time rather than flat stability; second, baseline psychological traits (the Big Five personality dimensions and generalized trust) systematically outperformed standard socioeconomic indicators (parental income, parental education) in predicting trajectory membership. However, as Chandler and Hachen acknowledged, their preliminary framework suffered from critical methodological limitations: it evaluated trajectories over a truncated six-wave horizon, relied on continuous Euclidean distance metrics that mischaracterize discrete count distributions, required ad-hoc single-imputation for missed waves, and treated total degree as an undifferentiated aggregate count.

This study builds upon and substantially unifies this line of inquiry by providing a principled, full-panel investigation of degree trajectories across the entire undergraduate life course. Tracking an undergraduate cohort across all eight collegiate semesters from freshman matriculation in August 2015 through senior graduation in May 2019 ($N = 457$), we formulate our analysis around three core questions. First, what is the overall empirical trajectory of personal network size across the complete collegiate life course, and does the widely observed contraction in total degree reflect an erosion of core supportive communities or an adaptive pruning of superficial campus ties? Second, when modeled using principled count-data mixture likelihoods, what distinct latent trajectory classes emerge inductively across the undergraduate career? Third, what baseline sociodemographic characteristics and psychological dispositions predict which trajectory group a student will follow?

To resolve these questions, we deploy a unified analytical framework across all eight waves. We first decompose total ego network degree into four functional relational layers—total alter nominations, affective closeness, contact frequency, and functional social support—demonstrating that while total degree contracts by roughly 24% over college, daily activated ties and close supportive circles remain exceptionally stable. Second, we implement Latent Class Growth Analysis (LCGA) with repeated-measures Poisson finite mixture models across the eight semesters, identifying three distinct developmental trajectory classes: Network Conservers, Moderate Winnowers, and Accelerated Winnowers. Third, we estimate multinomial logistic regression models to evaluate the predictive power of baseline sociodemographic background versus psychological dispositions. Finally, in an Appendix, we demonstrate that continuous multilevel Poisson growth curve models corroborate these findings, confirming that extraverts experience significantly steeper tie winnowing over time while generalized trust preserves overall network scale.

2 Theoretical Background and Hypotheses

2.1 Cognitive Constraints, Dunbar Layers, and Personal Network Dynamics

The structural scale of human sociality has long been theorized as bounded by biological and cognitive constraints. Dunbar (1992, 1998) social brain hypothesis posits that primate neocortical volume constrains the maximum number of stable interpersonal relationships an individual can monitor simultaneously, yielding a theoretical upper limit of roughly 150 personal relationships for humans. Within this outer boundary, egocentric networks exhibit a nested hierarchy of emotional closeness and contact frequency, typically organized into layers of approximately 5 support-clique intimates, 15 sympathy-group associates, 50 affinity-group contacts, and 150 total active acquaintances (Dunbar, 2018; Hill and Dunbar, 2003; Zhou et al., 2005).

In a recent extension of this framework, Saramäki et al. (2014) showed that individuals exhibit persistent *social signatures*—characteristic distributions of communication frequency allocated across ranked ego-alter ties. Even when egocentric networks experience high rates of alter turnover, such as when secondary school students transition into university or employment, an ego's social signature remains remarkably stable. Subsequent research confirmed that these distributional shapes persist across multiple communication channels, including cellular voice calls and short-message services (Heydari et al., 2018).

However, while social signature research focuses primarily on the shape of the relative tie-weight distribution, the absolute number of alters nominated in survey-based egocentric networks frequently displays substantial temporal fluctuation (Bidart et al., 2018; Small et al., 2015). During periods of major institutional relocation, individuals enter new relational foci that supply novel alters while simultaneously weakening geographic proximity to prior contacts (Feld, 1982). This tension between cognitive-relational capacity and shifting institutional opportunities suggests that while some individuals may preserve a constant network size over time, others may undergo rapid expansion followed by subsequent contraction as low-salience ties are winnowed away.

2.2 Psychological Dispositions vs. Sociodemographic Stratification

When individuals navigate open social contexts, such as a residential university campus, what factors govern whether their personal networks expand, contract, or remain stable? Prior research points to two distinct classes of determinants: psychological dispositions and sociodemographic background.

Psychological theories emphasize that personal network structure reflects broad individual differences in behavioral tendencies and interpersonal motivations, as captured by the Five Factor Model of personality (Costa and McCrae, 1992; Roberts et al., 2007). Extraversion, characterized by sociability, assertiveness, positive emotionality, and reward-seeking behavior, has consistently been linked to larger overall network size, more frequent social interactions, and higher rates of tie initiation (Asendorpf and Wilpers, 1998; Centellegher et al., 2017; Harris and Vazire, 2016). Highly extraverted individuals possess a higher motivational orientation toward social engagement, which may encourage exploratory tie accumulation. Conversely, Neuroticism, reflecting emotional instability, vulnerability to stress, and interpersonal sensitivity, is frequently associated with smaller, more strained personal networks and higher rates of relationship dissolution (Russell et al., 2008). Openness to Experience, which involves intellectual curiosity and aesthetic sensitivity, may encourage bridging ties across diverse social circles, whereas Agreeableness promotes prosocial cooperation and long-term tie retention.

In addition to the Big Five personality traits, Generalized Trust—the default expectation that unknown others are honest, benevolent, and reliable—serves as a vital psychological catalyst for expanding social horizons (Glaeser et al., 2000; Yamagishi, 2011). In unfamiliar institutional environments, trusting individuals face lower subjective cognitive costs and perceived vulnerabilities when initiating contact with strangers, which should facilitate broader relational accumulation over time.

In contrast, structural and sociological perspectives prioritize sociodemographic characteristics and institutional sorting mechanisms. Stratification researchers have long shown that social network access and capital are structured by gender identity, racial/ethnic categorization, and family socioeconomic status (Campbell and Lee, 1991; Lin, 2001; Marsden, 1987). Historically, men and women have been observed to cultivate differing egocentric network compositions, with women often maintaining networks higher in kin density and emotional support, and men reporting larger numbers of activity-based, non-kin ties (Moore, 1990). Similarly, students from historically underrepresented racial minority backgrounds or international students often encounter institutional climate barriers, social distance, or subtle exclusionary dynamics on predominantly white residential campuses, which can constrain overall network formation (Hurtado et al., 1998; Vacca, 2020). Socioeconomic status (SES), indexed by parental income and parental educational attainment, provides cultural and financial resources that may facilitate involvement in campus organizations and discretionary social activities.

Building on these perspectives, we formulate the following hypotheses:

Hypothesis 1 (Psychological Primacy): *Baseline psychological dispositions* *(the Big Five traits and Generalized Trust) will be stronger predictors of degree trajectory group membership than baseline parental socioeconomic background.*

Hypothesis 2 (Trust and Network Volume): *Higher baseline Generalized Trust will be positively associated with membership in the stable Conserver trajectory class rather than the winnowing trajectory classes.*

Hypothesis 3 (Accelerated Extravert Winnowing): *Extraverted individuals will cast a wider initial exploratory net upon entering college, but will experience significantly steeper rates of subsequent tie winnowing over time compared to introverted peers.*

2.3 Functional Decomposition: Tie Closeness and Supportive Relational Labor

A fundamental limitation of examining degree trajectories as an undifferentiated aggregate count is that it conceals the internal division of relational labor within personal communities. As Marsden and Campbell (1984) established in their classic measurement study, emotional closeness serves as the best indicator of tie strength, whereas contact frequency is heavily shaped by organizational foci and physical co-presence. Coworkers or dormmates may interact daily due to spatial proximity rather than mutual intimacy, whereas close kin or childhood friends may interact infrequently yet maintain profound affective depth (Granovetter, 1973; Lizardo, 2024; Smith, 2021).

Furthermore, social support exchanges are highly specialized across distinct relational domains (Wellman and Wortley, 1990). Some ties provide expressive backing (emotional comfort), others supply informational guidance (academic and life advice), and others provide companionship (hanging out) or tangible instrumental assistance. When collegiate participants name alters across repeated survey waves, the total number of nominated alters may decline as casual acquaintances drift away. However, if individuals actively protect their core supportive bonds, we should expect close, support-providing ties to remain stable even as total network size winnows down:

Hypothesis 4 (Functional Insulation): *The longitudinal decline in collegiate network size will be concentrated primarily among peripheral acquaintances, while close ties, daily activated ties, and support-providing ties will remain invariant across the eight collegiate waves.*

3 Data and Analytical Sample

3.1 The NetHealth Project

Data for this study come from the *NetHealth* study (https://sites.nd.edu/nethealth/), a longitudinal study tracking an entire undergraduate cohort at the University of Notre Dame from matriculation in August 2015 through senior graduation in May 2019 (Liu et al., 2018; Sepulvado et al., 2020; Wang et al., 2020). The *NetHealth* study was designed to investigate the reciprocal co-evolution of social network structures, health behaviors (physical activity, sleep patterns, stress), and academic performance. Participants were recruited prior to entering college and were surveyed repeatedly across their undergraduate careers, completing comprehensive baseline surveys alongside semester-by-semester egocentric network batteries.

3.2 Longitudinal Analytical Cohort

Our analytical cohort is defined across all eight waves of the study (Wave 1 in Fall 2015 to Wave 8 in Spring 2019). The longitudinal cohort consists of $N = 457$ undergraduate students who completed at least three full egocentric network survey waves across the four-year undergraduate timeline, completed the initial baseline attributes survey, and maintained active participation. For fully adjusted multinomial predictive models, complete baseline covariate data are available for $N = 433$ participants.

3.3 Variables

3.3.1 Ego Network Degree and Relational Layers

In each survey administration, participants completed an egocentric network generator that asked them to nominate their most important social contacts (friends, family members, romantic partners, university peers), with a maximum of 25 alters per wave. Total degree D_{it} for ego i at wave t is calculated as the total count of nominated alters ($D_{it} \in \{0, 1, \ldots, 25\}$). In addition, for each nominated alter, respondents provided detailed tie-level evaluations:

- **Affective Closeness:** Categorized as "Especially Close," "Merely Close," "Less than Close," or "Distant." Alters designated as especially close or merely close constitute the ego's strong/close tie subnetwork.

- **Contact Frequency:** Categorized as "Daily," "Weekly," "Monthly," or "Less than Monthly." Alters contacted on a daily basis define the daily activated tie layer.

- **Support-Providing Ties:** Binary indicators assessing whether the alter provided companionship ("hanging out"), academic or personal advice, emotional comfort, or financial assistance. Alters providing at least one form of support constitute the supportive subnetwork.

3.3.2 Baseline Psychological Traits

All independent variables were measured at baseline prior to or during Wave 1 matriculation, ensuring clear temporal ordering:

- **The Big Five Personality Inventory (BFI):** Standard validated multi-item scales measuring Extraversion, Neuroticism, Agreeableness, Conscientiousness, and Openness to Experience on five-point Likert scales (John and Srivastava, 1999). In analytical models, personality dimensions are standardized to z-scores ($\mu = 0, \sigma = 1$).

- **Generalized Trust Scale:** Measured using a validated multi-item instrument assessing general faith in humanity and expectations of benevolence, standardized to a z-score.

- **Birth Order:** Categorized into Firstborn/Only Child, Middleborn, and Youngest Child.

3.3.3 Sociodemographic Characteristics

We include a comprehensive battery of sociodemographic covariates measured at baseline to account for structural sorting and social background across the undergraduate cohort. Gender identity is coded as a binary indicator: women (1) and men (0). Racial and ethnic background is categorized as White, Asian, Black, Hispanic/Latino, and Other or International. Family socioeconomic status is captured through three distinct indicators: parental annual household income, measured on an eight-point ordinal scale ranging from less than $25,000 to $250,000 or more, alongside maternal and paternal educational attainment, each measured on five-point ordinal scales ranging from less than high school to graduate or professional degrees. In addition, we account for international status and language background using binary indicators for U.S. citizenship (1 = citizen, 0 = non-citizen) and primary home language (1 = English, 0 = non-English). Finally, reflecting the institutional setting of a historically faith-based private residential university, religious affiliation is categorized into Catholic, Protestant, other religious tradition, and no religious affiliation (atheist, agnostic, or none).

4 The Collegiate Network Landscape: Aggregate Evolution and Functional Decomposition

To resolve the substantive meaning of personal network evolution across college, we first examine the empirical trajectory of ego network size across all eight collegiate waves, extending from matriculation in August 2015 through senior graduation in May 2019 ($N = 457$). Table 1 displays the empirical means and standard errors for Total Degree, Close Ties, Daily Activated Ties, and Support-Providing Ties across all eight semesters.

Table 1: Longitudinal Means and Standard Errors of Decomposed Relational Dimensions Across Eight Collegiate Waves ($N = 457$)

Academic Wave	**Egos (N)**	**Total Degree (SE)**	**Close Ties (SE)**	**Daily Ties (SE)**	**Support Ties (SE)**
Wave 1 (Frosh Fall)	323	14.19 (0.28)	13.00 (0.27)	5.93 (0.19)	—
Wave 2 (Frosh Spr)	442	13.38 (0.28)	12.46 (0.27)	6.93 (0.19)	11.99 (0.31)
Wave 3 (Soph Fall)	379	12.27 (0.33)	11.48 (0.31)	5.36 (0.20)	12.03 (0.32)
Wave 4 (Soph Spr)	371	11.37 (0.32)	10.82 (0.30)	5.92 (0.19)	11.08 (0.31)
Wave 5 (Jun Fall)	348	11.13 (0.33)	10.56 (0.32)	5.40 (0.18)	10.64 (0.32)
Wave 6 (Jun Spr)	300	10.64 (0.36)	10.14 (0.34)	4.93 (0.21)	—
Wave 7 (Sen Fall)	254	10.88 (0.39)	10.28 (0.38)	4.76 (0.21)	10.40 (0.38)
Wave 8 (Sen Spr)	181	10.80 (0.45)	10.04 (0.44)	4.38 (0.26)	10.45 (0.45)

Figure 1 visualizes the multi-line trajectory profiles of these four relational dimensions across all eight collegiate semesters.

The empirical pattern displayed in Figure 1 provides decisive support for **Hypothesis 4** and fundamentally clarifies the nature of collegiate network contraction:

- **Total Ego Network Size** (blue line) exhibits a steady downward decline, falling from 14.19 alters in freshman fall (Wave 1) to 10.64 in junior spring (Wave 6) and stabilizing at 10.80 in senior spring (Wave 8), representing an overall net contraction of approximately 24%.

- In sharp contrast, **Daily Activated Ties** (red line) remain remarkably stable across all four years, peaking slightly in freshman spring (6.93 ties) before hovering consistently between 4.4 and 5.9 ties per wave across sophomore, junior, and senior years.

- Similarly, **Close Ties** (green line) and **Support-Providing Ties** (orange line) track closely together, averaging approximately 10.5 to 12.0 ties across college.

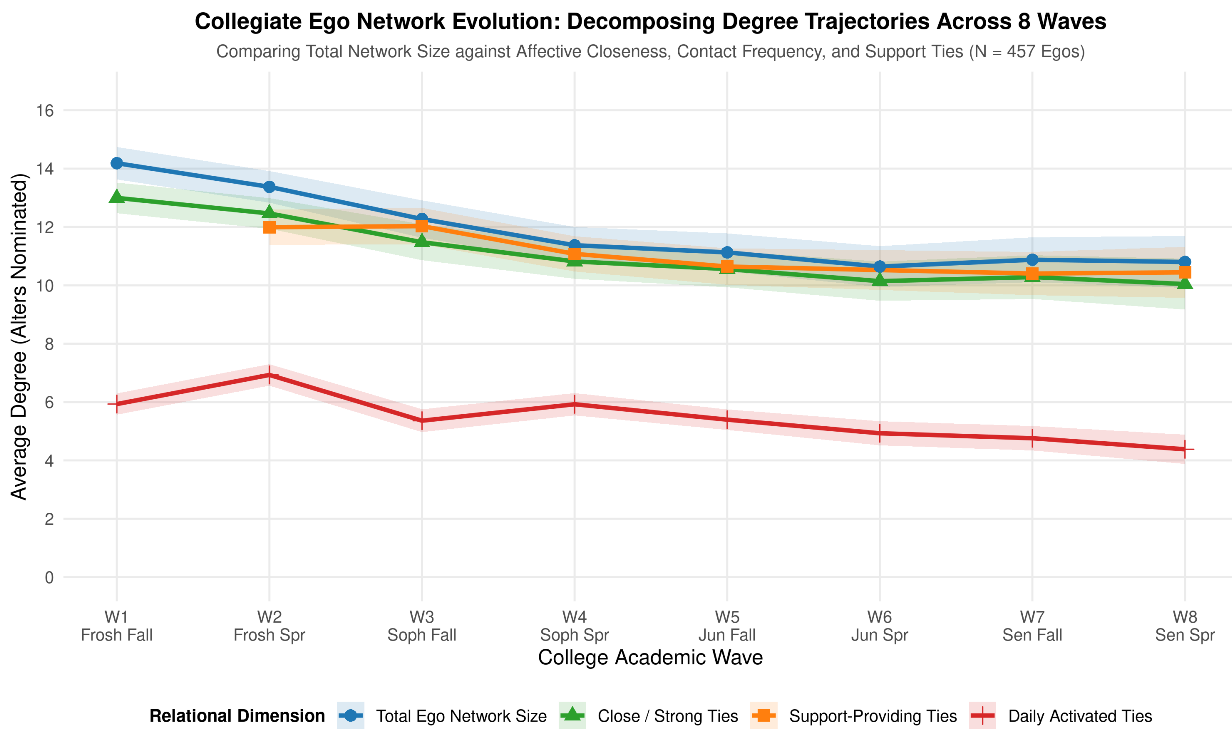

Figure 1: Decomposing ego network evolution across eight collegiate waves: Total Degree, Strong/Close Ties, Daily Activated Ties, and Support-Providing Ties ($N = 457$). Shaded ribbons indicate 95% confidence intervals.

This functional decomposition reveals that the apparent "downward degree trajectory" widely documented in aggregate student surveys is not an indicator of social isolation or an erosion of core supportive communities. Rather, students maintain a highly resilient, temporally stable inner core of daily activated associates (matching Dunbar's support clique of roughly 5 alters) and close, support-providing ties (matching the sympathy group of roughly 10 to 12 alters). The aggregate decline in network size is driven almost entirely by the winnowing of peripheral, low-salience acquaintances that were temporarily accumulated during the initial disorganization of freshman year. Once students establish enduring campus niches, they prune superficial connections to conserve cognitive and temporal bandwidth.

5 Discovering Latent Trajectory Classes via Poisson LCGA

5.1 Mixture Formulation and Model Selection

While aggregate trajectories reveal overall population trends, they mask substantial inter-individual heterogeneity in how students navigate relational winnowing. To identify distinct developmental pathways without imposing arbitrary boundary rules or Gaussian assumptions, we implement Latent Class Growth Analysis (LCGA) using repeated-measures Poisson finite mixture models (Grün and Leisch, 2008).

Specifying a Poisson emission distribution with linear and quadratic time terms centered at elapsed collegiate semester ($\text{Time}_{it} \in \{1, \dots, 8\}$), the conditional expectation of degree for ego i in latent class k is given by:

$$\log(\mathbb{E}[D_{it} \mid C_i = k]) = \beta_{0k} + \beta_{1k}\text{Time}_{it} + \beta_{2k}\text{Time}_{it}^2 \tag{1}$$

Observations are clustered at the ego level (|egoid), allowing the Expectation-Maximization (EM) algorithm to estimate class assignment probabilities while natively handling unbalanced panel observations across waves under missing-at-random assumptions. We evaluate candidate solutions from $K = 1$ to $K = 5$ latent classes.

Table 2 reports the information-theoretic fit statistics across candidate models.

As shown in Table 2, information criteria improve dramatically when moving from a single population trajectory ($K = 1$) to multi-class mixtures, with BIC dropping by 4,132.3 points from $K = 1$ to $K = 3$. While BIC continues to decline through $K = 5$, the marginal gains diminish substantially beyond $K = 3$. To balance mathematical parsimony with substantive interpretability, we select the three-class solution ($K = 3$) as the optimal representation of

Table 2: Latent Class Growth Analysis (LCGA) Model Fit Statistics Across Candidate Poisson Mixture Models on Eight-Wave Panel ($K = 1 \ldots 5, N = 457$)

Latent Classes	Log-Likelihood	Par	AIC	BIC	ΔBIC
$K = 1$	-9627.1	3	19260.3	19277.9	4364.4
$K = 2$	-7863.6	7	15741.2	15782.3	868.8
$K = 3$	-7529.6	11	15081.1	15145.6	232.2
$K = 4$	-7434.6	15	14899.1	14987.1	73.6
$K = 5$	-7382.0	19	14802.0	14913.4	0.0

collegiate trajectory morphology.

5.2 Latent Trajectory Profiles Across Eight Waves

Figure 2 displays the model-implied trajectories and observed ego sequences across all eight semesters for the optimal three-class solution.

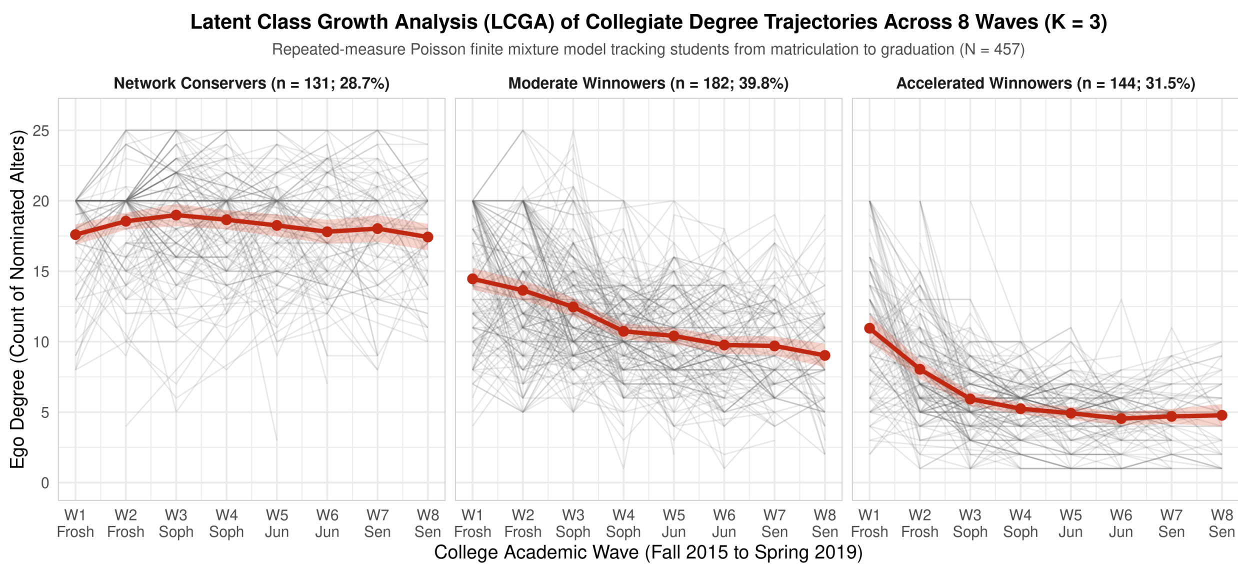

Figure 2: Latent Class Growth Analysis (LCGA) trajectory profiles from repeated-measures Poisson mixture model across eight collegiate waves ($K = 3, N = 457$). Thin grey lines denote individual ego trajectories; bold red lines and shaded ribbons indicate class mean trajectories and 95% confidence intervals.

The three latent classes capture distinct developmental archetypes of personal network evolution:

1. **Network Conservers ($n = 131$, 28.7%):** Students who matriculate with large personal networks ($\bar{D}_1 = 17.60$) and preserve an exceptionally stable network scale

throughout their entire undergraduate careers ($\bar{D}_8 = 17.43$), exhibiting essentially zero net winnowing over four years.

2. **Moderate Winnowers ($n = 182$, 39.8%):** The modal collegiate pathway, representing students who matriculate with average-sized circles ($\bar{D}_1 = 14.46$) and experience a steady, progressive winnowing across semesters, shedding roughly five to six alters by graduation ($\bar{D}_8 = 9.03$).

3. **Accelerated Winnowers ($n = 144$, 31.5%):** Students who enter college with somewhat smaller networks ($\bar{D}_1 = 10.95$), undergo an aggressive contraction across freshman and sophomore years, and stabilize at an intimate core of roughly 4 to 5 alters through senior graduation ($\bar{D}_8 = 4.78$).

5.3 Functional Tie Profiling Across Latent Classes

To ascertain whether winnowing classes experience an erosion of emotional closeness and social support, Figure 3 stratifies the functional tie decomposition (Total, Close, Daily, Support) across the three latent LCGA classes across all eight waves.

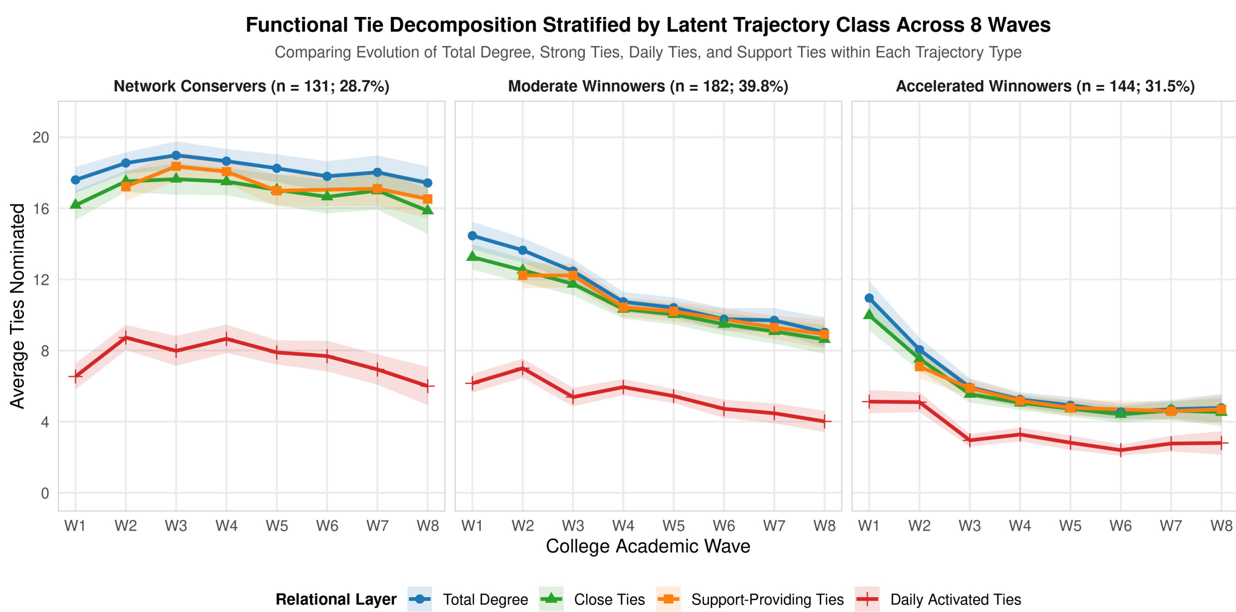

Figure 3: Functional tie decomposition (Total Degree, Close Ties, Daily Activated Ties, and Support-Providing Ties) stratified by latent trajectory class across all eight collegiate waves ($N = 457$). Shaded ribbons denote 95% confidence intervals.

Figure 3 provides compelling evidence regarding the internal mechanics of relational winnowing. For Network Conservers, high Total Degree ($\approx$ 17–19 alters) is mirrored by high

Close Ties ($\approx$ 16–17.5) and high Support-Providing Ties ($\approx$ 16.5–18.4), with daily activated ties remaining steady at roughly 7 to 9 alters. For Moderate Winnowers, Total Degree winnows from 14.5 down to 9.0 alters, but Close Ties (13.3 to 8.6) and Support Ties (12.2 to 8.9) track this contraction closely, while daily interaction stabilizes at roughly 4 alters. Most strikingly, for Accelerated Winnowers, the sharp contraction from 11.0 to 4.8 alters brings Total Degree, Close Ties, and Support Ties into perfect convergence around 4.5 to 5.0 alters by sophomore year. In other words, Accelerated Winnowers prune all peripheral acquaintances until their total network perfectly coincides with their core support clique.

6 Predicting Trajectory Group Membership: Multinomial Logistic Regressions

Having established the three latent trajectory classes across all eight collegiate waves, we examine whether baseline sociodemographic characteristics and psychological dispositions predict which developmental trajectory an individual follows. We estimate multinomial logistic regression models using `nnet::multinom` on the complete-case cohort ($N = 433$), setting the modal category—**Moderate Winnowers**—as the reference category.

Table 3 reports the multinomial logit parameter estimates, standard errors, Odds Ratios (Relative Risk Ratios), 95% confidence intervals, and p-values for both Network Conservers and Accelerated Winnowers relative to Moderate Winnowers.

Figure 4 visualizes the Odds Ratios and 95% confidence intervals in a publication-grade forest plot.

The multinomial regression results provide clear empirical evaluation of our core hypotheses:

- **Support for Hypothesis 1 (Psychological Primacy):** Classic indicators of family socioeconomic background—including parental income ($\text{OR} = 1.04, p = 0.601$ for Conservers; $\text{OR} = 0.99, p = 0.887$ for Winnowers) and maternal college degree ($\text{OR} = 0.93, p = 0.871$; $\text{OR} = 1.00, p = 0.993$)—exhibit zero predictive power. In contrast, psychological traits and trust display substantial substantive associations.

- **Support for Hypothesis 2 (Trust and Conserving Scale):** Higher baseline Generalized Trust increases the odds of belonging to the Network Conserver class by 31% per standard deviation ($\text{OR} = 1.31$, 95% CI: $[0.98, 1.76]$, $p = 0.072$), while reducing the likelihood of belonging to the Accelerated Winnower class ($\text{OR} = 0.87$). Trusting students are systematically more likely to sustain large, durable networks across all four years.

Table 3: Multinomial Logistic Regression Estimates Predicting Eight-Wave Latent Trajectory Class Membership (Reference Category: Moderate Winnowers; $N = 433$)

Predictor Variable	**Estimate (SE)**	**Odds Ratio [95% CI]**	***p*-value**
Panel A: Network Conservers (vs. Moderate Winnowers)			
Gender Identity: Woman (ref: Man)	0.36 (0.26)	1.43 [0.86, 2.38]	0.172
Race: Asian (ref: White)	-0.59 (0.50)	0.55 [0.21, 1.48]	0.238
Race: Black (ref: White)	-0.57 (0.65)	0.56 [0.16, 2.02]	0.380
Race: Hispanic/Latino (ref: White)	-0.32 (0.39)	0.72 [0.34, 1.56]	0.411
Race: Other/International (ref: White)	-0.27 (0.69)	0.77 [0.20, 2.94]	0.698
Parents' Income (Ordinal Scale)	0.04 (0.07)	1.04 [0.91, 1.18]	0.601
Mother College Degree (ref: Non-degree)	-0.07 (0.43)	0.93 [0.40, 2.18]	0.871
Extraversion (z-score)	-0.20 (0.13)	0.82 [0.63, 1.06]	0.131
Neuroticism (z-score)	0.01 (0.14)	1.01 [0.76, 1.33]	0.969
Agreeableness (z-score)	0.06 (0.15)	1.07 [0.80, 1.43]	0.663
Conscientiousness (z-score)	-0.06 (0.14)	0.94 [0.72, 1.24]	0.669
Openness (z-score)	0.12 (0.13)	1.13 [0.87, 1.46]	0.358
Generalized Trust (z-score)	0.27 (0.15)	1.31 [0.98, 1.76]	0.072
Panel B: Accelerated Winnowers (vs. Moderate Winnowers)			
Gender Identity: Woman (ref: Man)	0.04 (0.26)	1.04 [0.63, 1.72]	0.875
Race: Asian (ref: White)	0.41 (0.41)	1.51 [0.68, 3.35]	0.307
Race: Black (ref: White)	0.74 (0.49)	2.10 [0.80, 5.48]	0.131
Race: Hispanic/Latino (ref: White)	-0.36 (0.40)	0.70 [0.32, 1.52]	0.363
Race: Other/International (ref: White)	1.11 (0.53)	3.02 [1.07, 8.51]	0.037
Parents' Income (Ordinal Scale)	-0.01 (0.06)	0.99 [0.87, 1.12]	0.887
Mother College Degree (ref: Non-degree)	-0.00 (0.39)	1.00 [0.47, 2.13]	0.993
Extraversion (z-score)	-0.16 (0.13)	0.85 [0.66, 1.10]	0.210
Neuroticism (z-score)	-0.09 (0.14)	0.91 [0.69, 1.20]	0.517
Agreeableness (z-score)	-0.12 (0.14)	0.89 [0.68, 1.17]	0.395
Conscientiousness (z-score)	-0.09 (0.14)	0.91 [0.70, 1.19]	0.492
Openness (z-score)	0.19 (0.13)	1.21 [0.93, 1.56]	0.151
Generalized Trust (z-score)	-0.14 (0.14)	0.87 [0.66, 1.14]	0.305

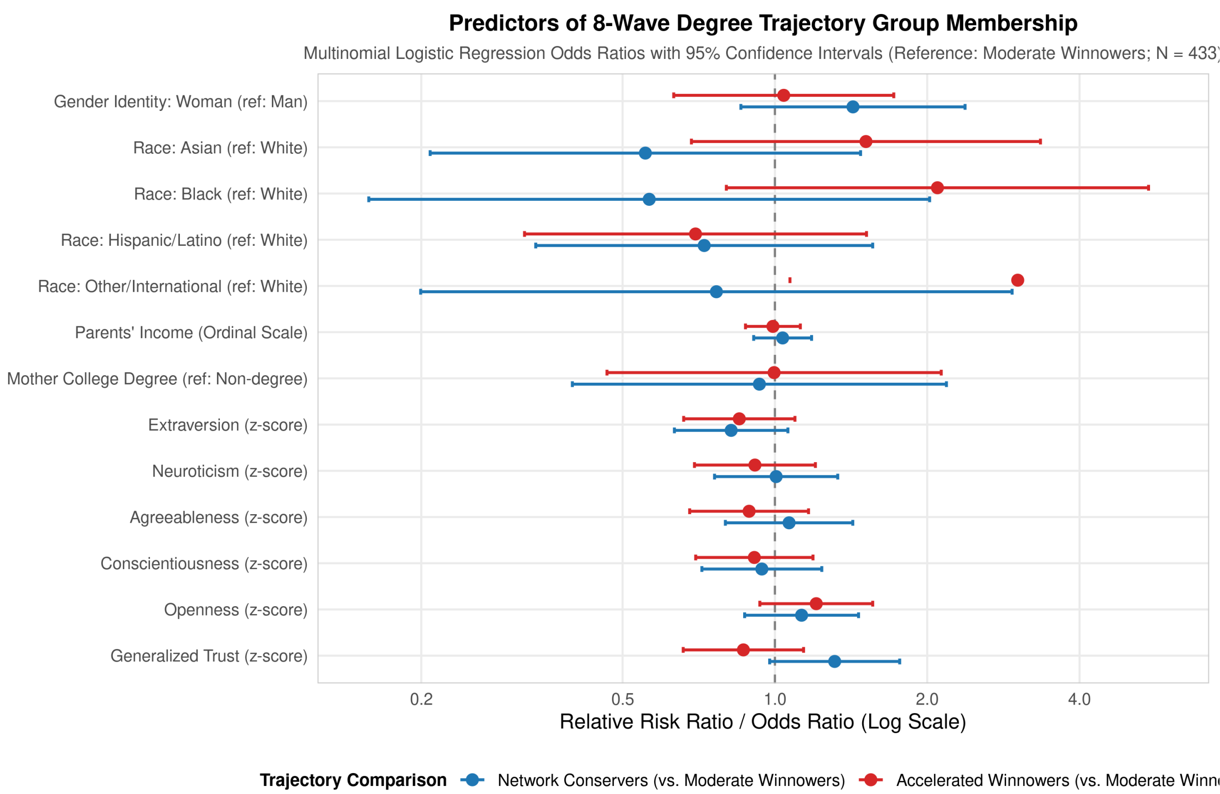

Figure 4: Forest plot of Odds Ratios (Relative Risk Ratios) from multivariable multinomial logistic regression predicting 8-wave latent trajectory class membership (Reference Category: Moderate Winnowers; $N = 433$). Error bars indicate 95% confidence intervals.

- **Race and Institutional Context:** Students identifying as Other or International face significantly higher odds of belonging to the Accelerated Winnower class (OR $= 3.02$, 95% CI: $[1.07, 8.51], p = 0.037$), reflecting institutional friction and social distance on a predominantly white residential campus.

7 Discussion and Conclusion

7.1 Summary of Key Results

Taken together, the empirical findings across the eight-wave panel provide a unified, comprehensive portrait of how ego network size evolves across the undergraduate life course. Rather than supporting a singular narrative of continuous growth or uniform relational decay, the results reveal an adaptive social architecture characterized by pervasive, timing-dependent tie winnowing alongside remarkable temporal stability in core supportive communities.

First, evaluating functional relational layers across all eight semesters reveals that the widely observed aggregate degree contraction is an artifact of counting total nominations without differentiating tie strength. While total degree shrinks by 24% between freshman matriculation and senior graduation (falling from 14.19 to 10.80 alters), daily activated ties (4.4 to 5.9 alters) and close, support-providing ties (10.5 to 12.0 alters) remain invariant across all four years. Students actively protect their intimate support clique and sympathy group while shedding peripheral campus acquaintances.

Second, Latent Class Growth Analysis with Poisson mixtures establishes that students do not follow a uniform winnowing trajectory. Instead, undergraduate personal communities divide into three distinct developmental archetypes: a stable cadre of Network Conservers ($n = 131$, 28.7%) who preserve large personal networks ($\approx$ 17.5–19.0 alters) throughout college; a modal group of Moderate Winnowers ($n = 182$, 39.8%) who matriculate with average-sized circles and steadily winnow down to ≈ 9 alters; and an Accelerated Winnower class ($n = 144$, 31.5%) that undergoes an early, rapid contraction to an intimate core of roughly 5 alters.

Third, multinomial logistic regressions confirm that trajectory group membership is governed primarily by baseline psychological dispositions and race rather than parental socioeconomic resources. Replicating the core insight of Chandler and Hachen (2018) within a unified full-panel mixture framework, parental income and parental education exhibit zero predictive capacity, whereas Generalized Trust promotes retention in the stable Conserver class, and international/minority status accelerates winnowing. Finally, continuous growth modeling in the Appendix confirms that extraverts experience significantly faster rates of tie winnowing over time ($\text{IRR} = 0.994, p = 0.032$), showing that extraversion accelerates early exploratory tie accumulation rather than buffering against subsequent tie attrition.

7.2 Limitations and Suggestions for Future Work

A central methodological challenge in longitudinal personal network research is operationalizing ego network boundaries and tie strength across repeated panel waves. In the NetHealth study, network degree was measured via free-recall name generators bounded at a maximum of 25 alters, followed by name interpreter batteries assessing contact frequency, emotional closeness, and supportive exchanges. While bounding nominations at 25 alters captures the active sympathy group and minimizes respondent fatigue, it imposes an artificial ceiling that may truncate the personal networks of exceptionally expansive egos, potentially muting initial baseline differences in early college. Future investigations can address this constraint by using unbounded roster-based nomination instruments or hybrid designs that integrate self-reported ego networks with objective behavioral trace data, such as continuous Bluetooth proximity sensing, smartphone call and text logs, and campus Wi-Fi co-location metrics (Centellegher et al., 2017; Sepulvado et al., 2020).

Second, because our analyses rely on observational panel data from a single collegiate cohort, parameter estimates must be interpreted as conditional longitudinal associations rather than strict causal effects. Several potential threats to causal identification warrant consideration. Although baseline psychological dispositions and sociodemographic characteristics were measured prior to or during the initial network wave, unmeasured contextual confounding remains possible. For example, residential hall assignments, campus club participation, and academic major sorting represent institutional foci that structure local opportunity pools (Feld, 1982; Small, 2009). If extraverted or trusting students systematically select into high-density dormitories or collaborative extracurricular activities, part of the estimated trajectory slopes may reflect organizational context rather than psychological traits per se. Future studies should leverage natural experiments in student residential placement, paired with detailed administrative records on campus involvement, to disentangle dispositional tendencies from institutional sorting mechanisms.

Third, our modeling framework cannot fully eliminate the possibility of reciprocal feedback loops between network dynamics and psychological well-being over time. While we treat personality traits as temporally antecedent, enduring individual dispositions (Costa and McCrae, 1992; John and Srivastava, 1999), personal network turnover can itself induce psychological stress, alter generalized trust, or reshape behavioral extraversion during formative life transitions (Asendorpf and Wilpers, 1998; Russell et al., 2008). Experiencing an acute contraction in social support during sophomore year may heighten neuroticism or erode generalized trust, generating a reinforcing spiral of relational withdrawal. Future research should implement continuous-time structural equation models, dynamic panel cross-lagged models, or latent curve models with structured residuals to trace the bidirectional co-evolution of personality traits and ego network structures across multiple years.

Fourth, several measurement granularity constraints of the survey design should be noted. Network surveys were administered once per semester (roughly two to three times per academic year). While this longitudinal rhythm effectively captures semester-to-semester macroscopic transitions, it cannot track the high-frequency micro-churn of tie activation, decay, and replacement that occurs across weeks or months within an academic term (Saramäki et al., 2014; Heydari et al., 2018). Furthermore, panel attrition across the four-year observation window reduced sample size in senior year (Wave 8, $n = 181$), reflecting survey fatigue and off-campus dispersal. Although our multilevel GLMMs and LCGA models naturally handle unbalanced panels under missing-at-random assumptions, future studies could employ mobile-app ecological momentary assessments (EMA) or monthly micro-surveys to capture high-resolution network churning while minimizing long-term attrition.

Finally, the empirical scope of this investigation is situated within a single, highly residential private university in the Midwestern United States characterized by a predominantly traditional-age, full-time undergraduate student body. The collegiate life course in this residential setting represents a unique institutional transition characterized by intense initial co-presence, communal living, and abrupt spatial reorganization upon graduation (Hurtado et al., 1998). Consequently, these findings cannot be assumed to generalize straightforwardly to commuter campuses, non-traditional student populations, working-class youth entering the labor market directly, or older adults navigating career changes or retirement. Future comparative research should extend this longitudinal trajectory framework to diverse institutional environments, working-adult cohorts, and cross-national contexts where personal community turnover is shaped by different organizational arrangements and life-course pacing (Bidart et al., 2018; Fischer, 1982).

7.3 Implications: The Adaptive Architecture of Personal Communities

Personal network analysis has long been animated by a foundational tension between structuralist social capital theories and evolutionary cognitive constraint models. In the classical social capital tradition (Granovetter, 1973; Lin, 2001; Borgatti et al., 2009), larger network degree is conceptualized as an unalloyed asset, providing access to diverse informational channels, instrumental sponsorship, and social support. From this perspective, the steady contraction of personal network size across college observed in empirical surveys might easily be interpreted as a problematic erosion of social integration or a deficit of relational connectivity. In contrast, the evolutionary cognitive constraint perspective pioneered by Dunbar (1992, 1998, 2018) posits that human sociality is biologically bounded by neocortical processing limits and finite temporal budgets, requiring individuals to distribute emotional and cognitive energy across nested hierarchical layers.

The empirical findings presented in this study challenge the pessimistic interpretation of network shrinkage by showing that longitudinal degree trajectories reflect an adaptive functional division of relational labor. Rather than signaling an undifferentiated decay of social ties, collegiate network contraction represents a deliberate, self-regulating process of relational winnowing. During the disorientation of freshman matriculation, students enter an unfamiliar social environment and temporarily cast a wide net, accumulating large volumes of peripheral, low-investment acquaintances. As academic demands intensify and enduring affinity groups coalesce, maintaining dozens of casual ties becomes cognitively and temporally unsustainable. Students resolve this constraint by winnowing peripheral ties while aggressively shielding their core supportive circles.

This functional insulation is clearly demonstrated by our eight-wave tie decomposition. Across all four undergraduate years, the volume of daily activated ties (averaging roughly 5 alters, precisely matching Dunbar's support clique) and close, support-providing ties (averaging 10 to 12 alters, aligning with the sympathy group) remains exceptionally stable, even as total degree contracts by more than 25%. What appears in aggregate survey counts as network decay is actually the elimination of redundant or superficial ties that contribute little to personal well-being or instrumental support. In this sense, collegiate network winnowing represents an adaptive optimization of personal communities, allowing individuals to concentrate finite social bandwidth where it yields the highest emotional and developmental returns.

Furthermore, our findings demonstrate that this winnowing process is fundamentally moderated by individual psychological dispositions. Personality traits serve as the underlying behavioral infrastructure that regulates how egos allocate relational labor and navigate institutional transition shocks. Extraverted students, driven by heightened reward sensitivity and exploratory social tendencies, experience the most aggressive network winnowing, transitioning from an expansive initial circle to a lean, selective collegiate network. Generalized trust, by contrast, lowers perceived transaction costs and interpersonal friction, enabling trusting individuals to sustain larger overall networks across time without sacrificing core supportive stability (Yamagishi, 2011).

Ultimately, these dynamics suggest that longitudinal degree trajectories reflect the structural stabilization of personal communities into persistent social signatures (Saramäki et al., 2014; Heydari et al., 2018). Rather than passive recipients of structural churn, individuals act as purposeful architects of their social worlds, using dispositional tendencies to explore available opportunity structures and winnowing peripheral connections to preserve an enduring core of supportive ties. By bridging formal latent growth mixtures, functional tie decomposition, and continuous multilevel modeling, this study provides a unified empirical foundation for understanding how personal networks adapt across major life transitions, opening new avenues for investigating the co-evolution of personality, social structure, and well-being across the adult life course.

Appendix: Continuous Multilevel Poisson Growth Curve Models (Robustness Check)

To verify that the discrete trajectory classes identified by LCGA reflect underlying continuous growth dynamics, we estimate multilevel Generalized Linear Mixed Models (GLMMs) with a Poisson response across all eight waves:

$$\log(\mathbb{E}[D_{it}]) = (\beta_0 + u_{0i}) + \beta_1 \mathrm{Time}_{it} + \mathbf{X}_i\boldsymbol{\beta} + (\mathrm{Time}_{it} \times \mathbf{Z}_i)\boldsymbol{\gamma} \quad (2)$$

where $u_{0i} \sim \mathcal{N}(0, \sigma_u^2)$ represents random ego intercepts capturing unobserved between-person heterogeneity, $\mathrm{Time}_{it} \in \{0, 1, \ldots, 7\}$ denotes elapsed semester centered at Wave 1 baseline, $\mathbf{X}_i$ is a vector of baseline sociodemographic and personality main effects, and $\mathbf{Z}_i$ captures cross-level interactions between focal personality traits (Extraversion and Neuroticism) and elapsed time.

Table A1 presents the model comparison fit hierarchy (Models 1–4) and the fully specified fixed effects estimates (Model 4).

Table A1: Fixed Effects Estimates from Multilevel Poisson Growth Curve GLMM with Random Ego Intercepts Across Eight Waves (Model 4; $N = 433$ egos, 2,456 observations)

Predictor Variable	**Estimate**	**Std. Error**	**IRR [95% CI]**	***p*-value**
Intercept (Baseline Degree)	2.53	0.07	12.55 [10.90, 14.44]	$< .001$
Time (Elapsed Collegiate Waves 0–7)	-0.06	0.00	0.95 [0.94, 0.95]	$< .001$
Extraversion (z-score)	0.02	0.02	1.02 [0.97, 1.07]	0.501
Neuroticism (z-score)	0.01	0.03	1.01 [0.96, 1.07]	0.609
Gender Identity: Woman (ref: Man)	0.01	0.04	1.01 [0.92, 1.10]	0.836
Race: Asian (ref: White)	-0.28	0.08	0.76 [0.65, 0.88]	$< .001$
Race: Black (ref: White)	-0.25	0.09	0.78 [0.65, 0.94]	0.008
Race: Hispanic/Latino (ref: White)	0.02	0.07	1.02 [0.89, 1.16]	0.773
Race: Other/International (ref: White)	-0.24	0.09	0.79 [0.66, 0.95]	0.011
Parents' Income (Ordinal Scale)	0.01	0.01	1.01 [0.99, 1.03]	0.325
Agreeableness (z-score)	0.04	0.02	1.04 [0.99, 1.09]	0.140
Conscientiousness (z-score)	0.00	0.02	1.00 [0.96, 1.05]	0.965
Openness (z-score)	-0.02	0.02	0.98 [0.94, 1.03]	0.442
Generalized Trust (z-score)	0.07	0.02	1.07 [1.02, 1.12]	0.006
Time $\times$ Extraversion	-0.01	0.00	0.99 [0.99, 1.00]	0.032
Time $\times$ Neuroticism	0.00	0.00	1.01 [1.00, 1.01]	0.101

Figure A1 visualizes the predicted degree trajectories across combinations of high versus low Extraversion (± 1 SD) and high versus low Neuroticism (± 1 SD).

The continuous growth estimates directly corroborate our primary mixture findings:

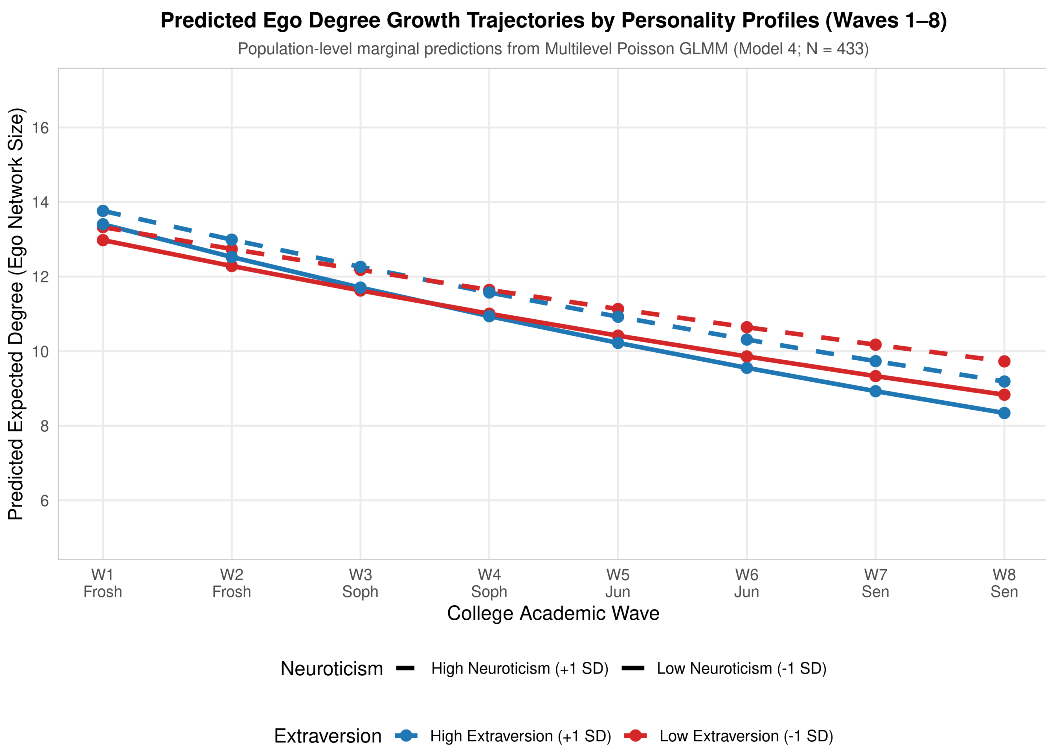

Figure A1: Predicted ego degree growth trajectories across eight collegiate waves by Extraversion ($\pm$1 SD) and Neuroticism ($\pm$1 SD) profiles estimated from Multilevel Poisson GLMM (Model 4; $N = 433$).

- On average, student ego networks contract by 5.5% per elapsed collegiate wave ($\text{IRR} = 0.945, p < 0.001$).

- Baseline Generalized Trust exerts a significant positive main effect on network scale ($\text{IRR} = 1.071, p = 0.006$), confirming that trusting students sustain larger personal communities throughout college.

- The interaction between Time and Extraversion is statistically significant and negative ($\text{IRR} = 0.994, p = 0.032$). As shown in Figure A1, extraverted students enter college with larger initial networks ($\approx$ 13.4–13.8 alters at Wave 1 vs. $\approx$ 12.9–13.3 for introverts) but undergo a faster rate of winnowing, crossing below their introverted peers by senior year ($\approx$ 8.3–9.2 alters vs. $\approx$ 8.8–9.7 alters).

References

Asendorpf, J. B. and Wilpers, S. (1998). Personality effects on social relationships. *Journal of Personality and Social Psychology*, 74(6):1531–1544.

Bidart, C., Degenne, A., and Grossetti, M. (2018). Personal networks typologies: A review of analytical approaches. *Social Networks*, 54:29–38.

Borgatti, S. P., Mehra, A., Brass, D. J., and Labianca, G. (2009). Network analysis in the social sciences. *Science*, 323(5916):892–895.

Campbell, K. E. and Lee, B. A. (1991). Name generators in surveys of personal networks. *Social Networks*, 13(3):203–221.

Centellegher, S., López, E., Saramäki, J., and Lepri, B. (2017). Personality traits and ego-network dynamics. *PLOS ONE*, 12(3):e0173110.

Chandler, M. J. and Hachen, D. (2018). Classifying and predicting degree trajectories in longitudinal ego networks. In *Sunbelt XXXVIII, International Network for Social Network Analysis (INSNA)*, Utrecht, Netherlands.

Costa, P. T. and McCrae, R. R. (1992). Four ways five factors are basic. *Personality and Individual Differences*, 13(6):653–665.

Dunbar, R. I. M. (1992). Neocortex size as a constraint on group size in primates. *Journal of Human Evolution*, 22(6):469–493.

Dunbar, R. I. M. (1998). The social brain hypothesis. *Evolutionary Anthropology*, 6(5):178–190.

Dunbar, R. I. M. (2018). The anatomy of friendship. *Trends in Cognitive Sciences*, 22(1):32–51.

Feld, S. L. (1982). Social structural determinants of similarity among associates. *American Sociological Review*, 47(6):797–801.

Fischer, C. S. (1982). *To Dwell Among Friends: Personal Networks in Town and City*. University of Chicago Press, Chicago, IL.

Glaeser, E. L., Laibson, D. I., Scheinkman, J. A., and Soutter, C. L. (2000). Measuring trust. *The Quarterly Journal of Economics*, 115(3):811–846.

Granovetter, M. S. (1973). The strength of weak ties. *American Journal of Sociology*, 78(6):1360–1380.

Grün, B. and Leisch, F. (2008). FlexMix version 2: Finite mixtures with concomitant variables and varying and constant parameters. *Journal of Statistical Software*, 28(4):1–35.

Harris, K. and Vazire, S. (2016). On friendship development and the Big Five personality traits. *Social and Personality Psychology Compass*, 10(11):647–667.

Heydari, S., Roberts, S. G. B., Dunbar, R. I. M., and Saramäki, J. (2018). Multichannel social signatures and persistent features of ego networks. *Applied Network Science*, 3(1):1–18.

Hill, R. A. and Dunbar, R. I. M. (2003). Social network size in humans. *Human Nature*, 14(1):53–72.

Hurtado, S., Milem, J., Clayton-Pedersen, A., and Allen, W. (1998). Enhancing campus climates for racial/ethnic diversity: Educational policy and practice. *The Review of Higher Education*, 21(3):279–302.

John, O. P. and Srivastava, S. (1999). The Big Five trait taxonomy: History, measurement, and theoretical perspectives. In Pervin, L. A. and John, O. P., editors, *Handbook of Personality: Theory and Research*, pages 102–138. Guilford Press, New York, NY.

Lin, N. (2001). *Social Capital: A Theory of Social Structure and Action*. Cambridge University Press, Cambridge, UK.

Liu, S., Hachen, D., Lizardo, O., Poellabauer, C., Striegel, A., and Milenković, T. (2018). Network analysis of the NetHealth data: Exploring co-evolution of individuals' social network positions and physical activities. *Applied Network Science*, 3(1):45.

Lizardo, O. (2024). Multi-frame relational sociology: Decoupling roles, sentiments, interactions, and exchanges in egocentric networks. *Sociological Theory*, 42(1):1–28.

Marsden, P. V. (1987). Core discussion networks of americans. *American Sociological Review*, 52(1):122–131.

Marsden, P. V. and Campbell, K. E. (1984). Measuring tie strength. *Social Forces*, 63(2):482–501.

McCarty, C., Lubbers, M. J., Vacca, R., and Molina, J. L. (2019). *Conducting Personal Network Research: A Practical Guide*. Guilford Press, New York, NY.

Moore, G. (1990). Structural determinants of men's and women's personal networks. *American Sociological Review*, 55(5):726–735.

Roberts, B. W., Kuncel, N. R., Shiner, R., Caspi, A., and Goldberg, L. R. (2007). The power of personality: The comparative validity of personality traits, socioeconomic status, and cognitive ability for predicting important life outcomes. *Perspectives on Psychological Science*, 2(4):313–345.

Russell, D. W., Booth, B., Reed, D., and Laughlin, P. R. (2008). Personality, social networks, and loneliness in later life. *Journal of Gerontology: Psychological Sciences*, 63(5):P313–P322.

Saramäki, J., Leicht, E. A., López, E., Roberts, S. G. B., Reed-Tsochas, F., and Dunbar, R. I. M. (2014). Persistence of social signatures in human communication. *Proceedings of the National Academy of Sciences*, 111(3):942–947.

Sepulvado, B., Wood, M., Wang, C., Fridmanski, E., Chandler, M., Lizardo, O., and Hachen, D. (2020). Predicting homophily and social network connectivity from dyadic behavioral similarity trajectory clusters. *Social Science Computer Review*, 40(1):186–205.

Small, M. L. (2009). *Unanticipated Gains: Origins of Network Inequality in Everyday Life*. Oxford University Press, Oxford, UK.

Small, M. L., Pamphile, V. D., and McMahan, P. (2015). How stable is the core discussion network? *Social Networks*, 40:90–102.

Smith, J. A. (2021). Social networks and social support. In Pescosolido, B. L., Martin, J. K., McLeod, J. D., and Rogers, A., editors, *Handbook of the Sociology of Mental Health*, pages 215–234. Springer, Cham, Switzerland.

Vacca, R. (2020). Structure in personal networks: Constructing and comparing typologies. *Network Science*, 8(2):245–269.

Wang, C., Lizardo, O., and Hachen, D. S. (2020). Neither influence nor selection: Examining co-evolution of political orientation and social networks in the NetSense and NetHealth studies. *PLOS ONE*, 15(5):e0233458.

Wellman, B. and Wortley, S. (1990). Different strokes from different folks: Community ties and social support. *American Journal of Sociology*, 96(3):558–588.

Yamagishi, T. (2011). *Trust: The Evolutionary Game of Mind and Society*. Springer, Tokyo, Japan.

Zhou, W.-X., Sornette, D., Hill, R. A., and Dunbar, R. I. M. (2005). Discrete hierarchical organization of social group sizes. *Proceedings of the Royal Society B: Biological Sciences*, 272(1561):439–444.